

React Events

What Are Events in React?

In React, events are actions triggered by users or the system — like clicking a button, typing in a field, submitting a form, hovering over an element, etc.

Step-by-Step Guide to React Events

Step 1: Basic Click Event

Let's start with the most common event: `onClick`.

```
import React from 'react';

function Button() {
  const handleClick = () => {
    alert('Button was clicked!');
  };

  return (
    <button onClick={handleClick}>
      Click Me!
    </button>
  );
}

export default Button;
```

Important Notes:

- We pass the function reference (`handleClick`) — not called (`handleClick()`).
- If you write `onClick={handleClick()}`, it runs immediately on render — ✗ wrong!
- `onClick={handleClick}` → passes function to be called later when clicked.

Step 2: Passing Arguments to Event Handlers

Sometimes you need to pass data to the handler — e.g., an ID or item name.

✗ Wrong Way:

```
<button onClick={handleClick(id)}> { /* This runs immediately! */ }
```

Correct Ways:

Option A: Arrow Function

```
const handleClick = (id) => {
  console.log("Clicked item:", id);
};
```

// In JSX:

```
<button onClick={() => handleClick(101)}>Delete Item</button>
```

Option B: Bind with `.bind()`

```
<button onClick={handleClick.bind(this, 101)}>Delete Item</button>
```

Step 3: Form Events — onChange, onSubmit

React handles form input changes via onChange, and submission via onSubmit.

```
import React, { useState } from 'react';
```

```
function TextInput() {
  const [text, setText] = useState("");

  const handleChange = (e) => {
    setText(e.target.value); // Update state as user types
  };

  return (
    <div>
      <input
        type="text"
        value={text}
        onChange={handleChange}
        placeholder="Type something..."
      />
      <p>You typed: {text}</p>
    </div>
  );
}
```

```
export default TextInput;
```

Other Common React Events

Event	Description	Example
onMouseEnter	Mouse enters element	<div onMouseEnter={handleHover}>Hover me</div>
onMouseLeave	Mouse leaves element	
onKeyDown / onKeyUp	Keyboard pressed/released	<input onKeyDown={handleKey} />
onFocus / onBlur	Element gains/loses focus	
onChange	Input value changed	Used in forms
onSubmit	Form submitted	
onDragStart	"onDrop	Drag & drop
onScroll	Scroll event	For infinite scroll